DU_Outliers at অলীকবচন (Olikbochon): Adaptive Multi-Signal LLM-Judge Ensembling with Verified Deterministic Retrieval for Bengali Hallucination Detection

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Abstract

We describe the DU_Outliers system for অলীকবচন (Olikbochon), the Bengali LLM Hallucination Detection Challenge at IUT 12th ICT Fest 2026. Given a Bengali prompt and a candidate response, with or without a supporting context passage, the task is to classify the response as faithful or hallucinated. Our system routes each instance into one of three prompting strategies by structural signal (**grounded**, **closed-book**, **arithmetic**), scores it with a single open-weight judge, Qwen2.5-32B-Instruct-AWQ, along two complementary axes – a calibrated single-token log-probability and a chain-of-thought self-consistency vote – and combines the two per route with weights selected by cross-validation between a linear blend and a logistic stacker. A deterministic, options-verified retrieval layer resolves a subset of closed-book instances without invoking the judge at all, and an adaptive compute allocator spends chain-of-thought votes only on instances the log-probability pass leaves genuinely uncertain, holding total inference cost roughly constant while sharpening the signal where it matters. On the organizers' held-out evaluation fold (5,000 instances), the system scores F_1=0.90216 on the hallucinated class, placing DU_Outliers **6th** among the finalist field. We report the full configuration, an honest account of what a small, non-representative calibration sample can and cannot certify, and the concrete engineering decisions – and reversions – that got us there.

Keywords: hallucination detection, LLM-as-judge, low-resource NLP, Bengali NLP, retrieval-augmented verification, self-consistency, chain-of-thought

1 Introduction

Large language models write fluent Bengali. They do not always write **true** Bengali. A model can correctly name the author of a poem when asked in English and misattribute it when asked in Bengali; it can narrate a historical event with confident, well-formed prose and the wrong year. These failures are dangerous precisely because they are fluent – nothing about the surface form signals unreliability to a reader who does not already know the answer. অলীকবচন (Olikbochon) ("the mirage of language") asks participants to build a detector for exactly this failure mode in Bengali, a language spoken by over 280 million people and structurally under-served by NLP research relative to that scale (Hasan et al., 2020).

The task spans three cultural-distance bands – globally stable facts, culturally situated Bangladesh-specific knowledge, and contested or recent facts – and mixes closed-book question answering with grounded, passage-conditioned verification and arithmetic reasoning. There is no conventional training set: a 299-row labeled sample supports local validation only, and the target failure mode concentrates in the culturally situated band, so a detector's closed-book world knowledge is tested precisely where general-purpose LLMs are weakest.

We treat this as an ensembling and calibration problem rather than a fine-tuning problem: a single strong open-weight judge in two complementary modes, combined with a deterministic verification layer over a curated Bengali knowledge bank, with effort spent on making blend weights, retrieval evidence, and compute budget honest and reproducible. Section 4 describes the pipeline; Section 5 reports cross-validated and held-out results; Section 6 states plainly where the system is weakest.

2 Related Work

Hallucination detection for LLM outputs has converged on a few families of method. **Self-consistency methods** sample multiple generations – often via chain-of-thought prompting (Wei et al., 2022; Wang et al., 2023) – and measure their mutual agreement (Manakul et al., 2023). **White-box methods** use token-level probabilities or log-probability curvature to flag hallucinated spans without extra sampling (Mitchell et al., 2023). **Factual-consistency methods** score a response against a source with a learned alignment function (Zha et al., 2023), the closest analogue to our grounded route, and retrieval-augmented approaches ground checks in retrieved evidence (Lewis et al., 2020) via rankers such as BM25 (Robertson and Zaragoza, 2009) or dense encoders (Wang et al., 2024). Ji et al. (2023) taxonomize these failures as intrinsic versus extrinsic, a split that maps onto our grounded/closed-book routing. Our `lp32`/`cot32` pair combines the first two families but fits their blend per route.

Almost all of this literature is developed and evaluated in English. Vu et al. (2024) show state-of-the-art LLM fact-checkers degrade substantially on low-resource languages, precisely the regime অলীকবচন tests. Bengali NLP resources remain scarce: Bhattacharjee et al. (2022) introduce BanglaBERT, and cross-lingual encoders such as XLM-R (Conneau et al., 2020) and multilingual BERT (Devlin et al., 2019) remain common baselines because monolingual alternatives are limited. Our system departs by not fine-tuning a classifier at all; it treats a general-purpose LLM as a zero-shot judge and fine-tunes nothing except a discarded LoRA verifier (Hu et al., 2022) (Section 6), sidestepping the need for a large labeled Bengali hallucination corpus that does not yet exist.

3 Task and Data

Labels and routing. Each instance consists of a Bengali prompt, a candidate response, and optionally a supporting context passage; the label is binary (faithful/hallucinated), scored as F_1 on the hallucinated class. We route each instance via cheap structural signals rather than a learned router: prompts matching an arithmetic/word-problem pattern go to **math**; remaining instances with a context passage go to **grounded**; the rest are **closed-book**. On the 5,000-row held-out fold this yields 2,340 grounded (46.8%), 2,387 closed-book (47.7%), and 273 math (5.5%) instances – closed-book and grounded jointly dominate, and closed-book is also the least-protected route (Section 6).

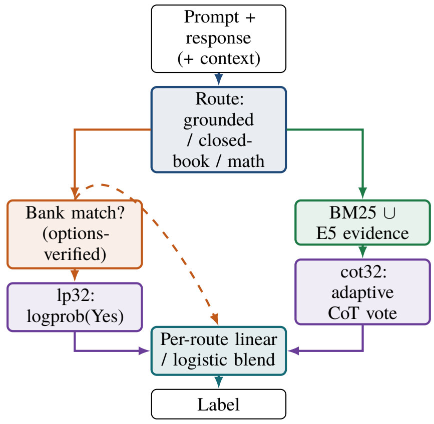

Figure 1: Pipeline overview. A bank match with response-in-options resolves closed-book instances directly (dashed path skips the judge); everything else is scored along two signals and combined per route by a cross-validated linear or logistic blend.

Calibration sample. A 299-row labeled sample is provided for local validation. A naive 5-fold split leaves roughly 4–5 examples per fold for the math route alone, and the sample does not claim to be representative of the held-out distribution's difficulty mix. We treat this as a structural constraint, not a solvable bug: cross-validated numbers in Section 5 are optimistic, lower-variance estimates, not predictions of held-out performance.

External data. All external data is public, declared, and disjoint from the competition test set: Bengali Wikipedia (`wikimedia/wikipedia`, config `20231101.bn`, CC BY-SA) rebuilt as a passage corpus, a supplementary Wikipedia mirror, a Bengali QA corpus, and a curated question–answer bank from public Bangladesh Civil Service exam archives, `titulm-bangla-mmlu`, and a Bangla idiom resource. Separately, we score a source-grouped, class-balanced 2,496-row subsample of the organizer-provided BanglaHalluEval benchmark and blend it into the calibration fit (Section 4.5) – never into the test set or the cross-validated score reported as "honest."

4 Method

Figure 1 shows the full pipeline. Every instance passes through routing, then (for closed-book instances) a deterministic verification layer, then judge scoring along two signals, and finally a per-route learned combination.

4.1 Judge and Signals

The judge is Qwen2.5-32B-Instruct-AWQ (Qwen Team, 2024), served locally with vLLM (Kwon et al., 2023) under AWQ 4-bit quantization (Lin et al., 2024) across two GPUs, with a 14B AWQ variant as an automatic fallback if the 32B snapshot fails to load. Every instance is scored along two signals:

lp32 is a single-token verification probability: the judge is asked to answer "Yes" or "No" to whether the response is faithful (grounded route) or factually correct (closed-book route), and we renormalize the softmax probability mass over the Yes/No token set to obtain a calibrated scalar in $[0, 1]$, following log-probability verification as in white-box hallucination detection (Mitchell et al., 2023).

cot32 is a chain-of-thought self-consistency vote (Manakul et al., 2023): the judge is asked to reason briefly and then emit an explicit verdict, sampled k times at temperature 0.7, and the signal is the fraction of samples voting "faithful." For the math route, which has no natural yes/no framing, the judge instead solves the problem independently and compares its own answer to the proposed one; here `lp32` and `cot32` coincide by construction.

Both prompts include an explicit trap rule: copying passage words does not make a response faithful if it answers the wrong slot (wrong entity, date, or number).

4.2 Deterministic Bank Verification

Closed-book instances are first checked against a BM25 index built exclusively over question–answer bank passages (144,701 passages on the held-out run), requiring $\geq$ 0.8 token coverage of the query question against the bank question. When a bank passage carries an explicit option list, we require the test response to literally appear among the matched question's options before trusting the match; if not, we conclude the wrong question was matched (a template collision, e.g. two "Who wrote X?" questions sharing most tokens) and defer to the judge. An earlier, symmetric configuration without this check – any mismatch against a covered bank question scored hallucinated – produced a real leaderboard regression from 0.858 to 0.838, traced to several hundred template-collision false positives. Without an option list, a token match is evidence of faithfulness; a mismatch is evidence of hallucination only when bank and test questions are near-identical (Jaccard $\geq$ 0.95). On the held-out run this layer resolves 40 of 2,387 closed-book instances (1.7%) without a judge call.

4.3 Retrieval

Grounded instances use their provided context directly. Closed-book instances additionally receive retrieved evidence: the top-4 BM25 passages over the union of all attached Wikipedia and bank corpora (616,065 passages), unioned with the top-4 by cosine similarity under a multilingual E5 sentence encoder (Wang et al., 2024). The semantic index is cached as a passage-embedding matrix, but a cached file is only trusted if a SHA-1 fingerprint of the full corpus text matches the fingerprint stored alongside it: a plain row-count check cannot detect a corpus rebuilt in a different order at the same size, and a stale cache passing that weaker check previously produced a 0.851 leaderboard score from silently-stale embeddings that were, in effect, pure BM25 retrieval. On the held-out run the fingerprint correctly rejected a reordered, larger corpus and recomputed all 616,065 embeddings in 11 minutes.

4.4 Adaptive Compute Allocation

Rather than sampling a fixed number of chain-of-thought votes for every instance in a route, we first compute `lp32` and skip the chain-of-thought pass entirely for any instance where it already reads confidently ($\leq$ 0.05 or $\geq$ 0.95), setting `cot32 := lp32` for that instance. The judge calls saved are redirected to the instances that remain: those get $k_{\text{base}}+2$ votes instead of k_{base}. On the held-out run, 2,263 of 2,340 grounded instances (96.7%) and 2,178 of 2,347 remaining closed-book instances (92.8%) were resolved by `lp32` alone, concentrating chain-of-thought sampling on the 77 and 169 instances, respectively, that the log-probability pass could not confidently decide.

4.5 Per-Route Calibration

For each route, we fit a combination of `lp32` and `cot32` and a decision threshold on labeled data via 5-fold cross-validation on the 299-row calibration sample, then re-fit on the full sample augmented with the scored BanglaHalluEval subsample restricted to matching routes (grounded: 1,196 rows; closed-book: 600 rows; math: 700 rows). Two combination families are compared inside each cross-validation fold: a linear blend, $s = w_{lp} \cdot$ lp32 $+ w_{cot} \cdot$ cot32 with (w_{lp}, w_{cot}) and a threshold chosen by grid search, and a small L_2-regularized

Route	n (train+ext)	Weights (lp,cot)	CV F_1
Grounded	1,322	0.8, 0.2	0.845
Closed-book	748	0.6, 0.4	0.669
Math	721	0.0, 1.0	0.922

Table 1: Final per-route blend weights and in-sample macro F_1 after augmenting the 299-row calibration sample with the matching BanglaHalluEval subsample. All three routes selected the linear blend over the logistic stacker on cross-validation.

logistic regression over the same two features. The family that wins on **held-out** fold performance – never on in-sample fit – is used for that route's final model, so a more flexible model is adopted only where it demonstrably generalizes better.

5 Results

On the honest, competition-data-only 5-fold cross-validation (299 rows, no external augmentation), the system reaches macro F_1=0.868 and hallucinated-class F_1=0.867. Table 1 shows the final per-route weights fit on the augmented calibration pool. Two patterns stand out. First, the grounded route weights `lp32` four times as heavily as `cot32` (0.8 vs. 0.2): a single calibrated verification probability already separates faithful from hallucinated responses well when a context passage is present, and the additional chain-of-thought pass contributes comparatively little marginal signal. Second, the math route collapses to pure `cot32` (weight 1.0): this is expected, since `lp32` and `cot32` are defined identically for math (the judge solving the problem itself), so the fitted weight is not meaningfully choosing between two different signals so much as confirming they carry the same information. The logistic stacker matched or slightly trailed the linear blend on every route in cross-validation (grounded: 0.837 vs. 0.840; closed-book: 0.867 vs. 0.867, tied), so the linear blend was retained everywhere.

On the organizers' held-out evaluation fold (5,000 instances, disjoint from both the public and private Kaggle test splits), the final submission scores F_1=0.90216 on the hallucinated class, placing DU_Outliers **6th** in the finalist field and securing an onsite slot. The predicted label distribution is 55.9% hallucinated / 44.1% faithful overall, with mean predicted-faithful rates of 0.361 (grounded), 0.487 (closed-book), and 0.718 (math) per route – broadly consistent with the calibration sample's known class balance and providing no evidence of a collapsed route.

6 Discussion and Limitations

We state these plainly, since they were load-bearing decisions during development, not afterthoughts.

The calibration sample does not fully represent the test distribution. The 299-row sample is an order of magnitude smaller than the held-out fold. Earlier regressions arose from failure modes invisible in 299-row cross-validation – notably the template-collision pattern Section 4.2 closes. BanglaHalluEval augmentation widens the labeled pool without touching the honest cross-validated score, but cannot manufacture representativeness the original sample lacks.

Closed-book is the least-protected route. With no bank match and no ground-truth passage, a closed-book prediction is only as good as the judge's own world knowledge plus retrieved evidence, and this is precisely the route where the competition's culturally-situated band concentrates. Its cross-validated F_1 (0.669) is visibly lower than grounded (0.845) or math (0.922) in Table 1, which we read as an honest signal of route difficulty, not a tuning failure.

A LoRA-based verifier signal was built and discarded. A LoRA adapter (Hu et al., 2022) was fine-tuned as a third signal, but on the same 299 rows later used for calibration; any weight learned from its output was leakage, and two submissions with it enabled scored 0.767 and 0.806, both below the no-adapter baseline. We keep it disabled, since retraining on disjoint data remains future work.

Single-judge dependence. The system depends on one base model's calibration quality; a second judge is not part of the ensemble.

7 Conclusion

DU_Outliers treats Bengali hallucination detection as a calibration problem over a single open-weight judge, not a fine-tuning problem requiring a Bengali corpus that does not exist. Structural routing, verified retrieval, adaptive compute, and cross-validated calibration reach F_1=0.90216 on the held-out fold, placing the team **6th** in the finalist field, with a documented history of what did **not** work, each diagnosed from real leaderboard evidence rather than assumed a priori.

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